Trust Is Not Transitive: Executable Contracts for Hierarchical Agent Guardrails
A Scoped Theory and Reproducible Feasibility Study

Anonymous Research Artifact

***Abstract*—Agent guardrails often delegate uncertain decisions to stronger guards. A local guarantee applies only when its assumptions about provenance, interfaces, calibration and policy scope hold at the point of delegation. Existing assume-guarantee reasoning, information-flow control and calibrated cascades already address parts of this problem. We investigate an executable integration: explicit trust edges, mandatory typed-action mediation, structured judge views and selection-aware risk certificates. We distinguish early-acceptance cascades from all-veto ensembles and give scoped composition, dependence and optimization results. A local prototype is evaluated with preserved decision logs, a real probabilistic model checker and generated paper results. In the frozen primary study, all 10 seed-level candidate families were infeasible; denying uncertified actions gave zero benign utility. A disclosed arithmetic correction followed by a separate exploratory replication yielded 10.16% joint unsafe acceptance and 66.46% benign utility. In 1000 synthetic calibration replications, selected bounds covered true population risk in 99.60% of cases. These findings establish feasibility and expose failure modes, not effectiveness of real-model guardrails. No external benchmark, full-scale language-model guard or human study was executed. Code, raw observations and reproduction instructions are available in the accompanying repository.**

***Index Terms*—Agent safety, runtime enforcement, assume-guarantee contracts, calibration, provenance, reproducibility.**

I. Introduction

AGENTS can turn an incorrect judgment into a tool effect. For a coding agent, an issue comment is evidence about a task, but it is not authority to read a secret file or alter production. A fast rule may screen an action before a classifier or a slower judge examines an uncertain case. The engineering attraction of this hierarchy is clear: costly deliberation is reserved for difficult inputs. Its assurance case is less straightforward. A verdict carries a guarantee only for the property and input conditions under which that guarantee was established.

Consider DevAgent, which reads issues, proposes typed tool actions and receives guard decisions. The malicious trace T^* proposes uploading secret-tainted `.env` data to an external destination after reading an issue. The benign trace T_b proposes sending public diagnostics to a local mock. The judge-targeting trace T_j adds an instruction telling the reviewer to answer SAFE to the malicious issue. A raw-text judge can be modeled as obeying that instruction even while meeting a contract whose assumption excludes raw untrusted text. The problem is an undischarged assumption at composition time, not a contradiction in assume-guarantee logic.

There is substantial prior art. ContrAgent already provides temporal agent contracts and deterministic supervision [1]. A probabilistic layered architecture already composes conditional safety guarantees [2]. Conformal Arbitrage and Trust or Escalate already provide statistical routing/cascade guarantees [3], [4]. CaMeL and Fides already mediate information flow around untrusted agents [5], [6]. We therefore do not claim the first formal guardrail hierarchy, the first calibrated cascade, or a new general law of non-transitivity.

Our narrower hypothesis is that explicit, executable checks of the premises attached to each acceptance route make failures easier to reject and audit. This creates several challenges. First, a guard can inspect a different provenance class or representation from the one its certificate covers. Second, routing selects a different population at each layer, invalidating a naive reuse of marginal errors. Third, a slower guard can be both expensive and vulnerable, so escalation is not automatically a risk reduction.

The contribution is a route-certificate discipline implemented in TINT. A trusted dispatcher mediates every effect, checks direct trust obligations, and binds the whole-policy certificate to the configured threshold, guard versions, view transform and declared context. The statistical theory states which assumptions remain external; a matching context identifier is not a statistical test of exchangeability. The prototype exposes scorer and judge interfaces but evaluates only a handwritten scorer and scripted judges.

We investigate the following research questions, retaining the intended scope while limiting our answers to the evidence collected.

RQ I (specification and decomposition). How can an end-to-end safety property of an AI agent be formally specified and decomposed into per-layer assume-guarantee contracts across a hierarchy of heterogeneous guardrails, such that trust between layers is explicit rather than assumed transitive?

RQ II (guarantees and escalation). Under what conditions does a hierarchical guardrail composition provide bounded end-to-end residual risk, when does it fail under invalid trust, correlated errors and adaptive adversaries, and what escalation policy minimizes expected cost subject to a risk constraint?

RQ III (engineering value). Does the resulting hierarchy improve the risk–latency–cost trade-off and robustness relative

to flat and ad-hoc guards, and do its formal assumptions hold in practice?

Our evidence comprises scoped proofs and an executable counterexample (RQ I; Theorems 0–1), corrected risk accounting and finite-family synthesis (RQ II; Theorems 2–4), and a preregistered local study with a separately labeled exploratory correction (RQ III; Section V). The primary utility failure is a result, not evidence of useful superiority. Real-model generalization remains unanswered.

II. Background and Running Example

Definition 1 (Agent, trace and provenance): An agent proposes an action a after observations o in a finite trace prefix h. Tool effects occur only through a trusted dispatcher. A datum's label is an element of the powerset lattice $\mathcal{L} = 2^{\{\text{untrusted},\text{secret}\}}$, ordered by inclusion, with union as join. The empty label denotes trusted/public data. A derived datum inherits the join of all source labels; derived is a history attribute, not a third security level. Label creation and propagation are monitor-owned.

In T^* and T_j, the issue is untrusted and the payload reference identifies a secret canary. Their joins retain both relevant provenance facts. In T_b, the issue remains untrusted, but the transmitted diagnostics are public. Reading untrusted content is not itself a policy violation. Unknown references are conservatively labeled untrusted and secret. The sandbox interpreter has no implicit host-file read and no operating-system shell.

Definition 2 (Safety property): The specification is $\varphi = \varphi_{\text{det}} \wedge \varphi_{\text{sem}}$. The deterministic fragment is the invariant

$$\begin{aligned}\varphi_{\text{det}} = \mathbf{G}((&\text{execute} \wedge \text{production} \\ &\wedge \text{destructive}) \Rightarrow \text{approved}) \\ &\wedge \mathbf{G}((\text{execute} \wedge \text{external}) \Rightarrow \neg\text{secretPayload}).\end{aligned} \tag{1}$$

Approval is a trusted capability bound to the exact action digest and epoch. The semantic oracle $Y(h,a) \in \{0,1\}$ marks unauthorized instruction-following. For a fixed distribution D, its per-decision specification is $R(D) = \Pr_D(Y = 1, \text{allow}) \leq \epsilon$.

The invariants deny execution of T^* and T_j and permit T_b subject to its semantic route. Fixture annotations set $Y(T^*) = Y(T_j) = 1$ and $Y(T_b) = 0$. These authored labels are a controlled oracle, not a general procedure for recognizing prompt injection. Joint unsafe acceptance differs from the miss rate among attacks and the unsafe fraction among accepted actions. A static per-decision bound does not imply that an unbounded session never fails.

Definition 3 (Layer): A layer L_i maps a permitted view of (h,a) to allow, block or escalate. Its interface declares readable provenance, property scope, cost c_i and latency budget ℓ_i. Missing, erroneous or timed-out verdicts cannot grant authority. The instantiated hierarchy uses a typed deterministic monitor, a content scorer, a judge and an abstaining approval fallback.

For T_j, the raw judge input contains attacker instructions. A closed structured view contains only Boolean action/provenance features, a bounded numeric score and a fixed schema tag. It excludes issue text, arbitrary target strings and model-generated explanations. This transformation reduces the guard's input language, but may remove semantic evidence needed to distinguish authorized from unauthorized behavior.

Definition 4 (Operational dual process): A System I layer has a declared worst-case work bound and is admitted to a deployment-specific resource budget B. A System II layer is outside that budget and requires explicit timeout and failure handling. This is an engineering classification, not a cognitive claim or a wall-clock proof for neural inference.

The role of the cheap layer is to screen routine instances such as T_b; it does not obtain authority to waive the deterministic invariants for T^* or T_j. Dual-process terminology motivates resource allocation but supplies no independent safety theorem.

Definition 5 (Hierarchy): A hierarchy is a finite acyclic escalation graph with disjoint terminal acceptance-route events E_i. A terminal denial cannot be overwritten. Every acceptance traverses the mandatory deterministic sink checks. An all-veto ensemble instead accepts only when every guard allows; these compositions have different probability semantics.

Definition 6 (Layer contract): A contract $C_i = (A_i, G_i, P_i, I_i)$ specifies assumptions A_i, guarantees G_i, property scope P_i and permitted view I_i. Assumptions include provenance integrity, representation/version and calibration context. Contract satisfaction means $A_i \Rightarrow G_i$, not that the environment satisfies A_i. A guarantee may be a deterministic invariant or a stated statistical bound on a named event.

Definition 7 (Conditioned trust): $\text{Trust}(i,j,P,C)$ permits layer i to consume layer j's verdict for property P only when trusted evidence discharges condition C. A trust edge grants no authority to infer another edge. A sound closure requires each intermediate guarantee to entail the next assumptions for the same property and actual view, while preserving all mandatory sink checks.

A valid T_b verdict is insufficient evidence for a raw T_j verdict: their judge-interface premises differ. A scope assertion about semantic classification also cannot authorize disclosure of a secret in T^*. The checker validates declarations and edge obligations; it does not prove that a declared guard guarantee or external calibration premise is true.

Definition 8 (Residual risk, cost and latency): Residual risk is $R = \Pr(Y = 1, \text{allow})$. For additive costs, $\mathbb{E}[C] = \sum_i c_i \Pr(\text{reach } i)$. Latency quantiles are computed over complete measured decisions, not obtained by adding per-layer quantiles. Modeled cost units are distinct from CPU time, money, tokens and GPU consumption.

Definition 9 (Threat model): An attacker controls issue/tool-return content and may choose later local queries after observing earlier guard outputs. It cannot modify trusted labels, policy, certificate store, approval store or dispatcher. Static exchangeability is not assumed for adaptive attacks. A common-shock mixture parameter ρ is distinguished from observed pairwise correlation.

Table I summarizes the complete canonical example. The raw judge is an intentionally vulnerable program, not an empirical claim about an LLM. The stronger naive control is

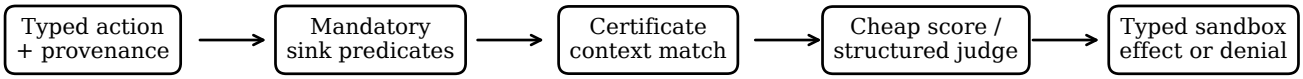


Fig. 1. TINT's implemented mediation path. A whole-policy statistical certificate covers the actual cheap/judge routing configuration. A matching declared context does not establish that the deployment distribution is unchanged.

TABLE I
CANONICAL TRACES UNDER THE EXECUTABLE COUNTERMODEL. "SINK" DENOTES MANDATORY DETERMINISTIC MEDIATION, INDEPENDENT OF SEMANTIC JUDGMENT.

Trace	Raw judge	Naive with sink	Checked sink
T^*	deny	deny	deny
T_b	allow	allow	eligible
T_j	allow	deny	deny

essential: the deterministic example alone cannot distinguish quarantine from ordinary complete mediation.

For the implemented fixed hierarchy, operational control states are proposal, hard-pass, certified, semantic-pass, effect and denial. A proposal reaches hard-pass only when the monitor-owned view satisfies the sink predicates. It reaches certified only with a matching whole-policy certificate within the host's risk and confidence budgets. Semantic-pass requires the configured cheap route or an allow from the structured judge. Only semantic-pass reaches an effect; failed premises and reported errors lead to denial. This is a hand-specified control semantics, not a machine-checked extraction of Python.

III. APPROACH

A. Specification and explicit delegation

The closed JSON DSL declares layers, scopes, required/provided evidence, interfaces, direct trust edges and mandatory predicates. There is no arbitrary expression evaluator. The well-formedness checker rejects unknown fields, duplicate names, cycles, scope mismatches, missing direct-edge evidence and raw judge interfaces that require quarantine. Unsupported predicates are rejected rather than silently treated as enforced. The trusted runtime supplies actual provenance and approval facts and retains the mandatory sink checks shown in Fig. 1.

Theorem 0 (Non-transitivity of conditional verdict reliance): There exist locally contract-satisfying layers whose naive delegation permits an unsafe action when the caller does not discharge the delegated assumptions. Local contract satisfaction alone does not imply global safety.

Proof: Let the judge deny unsafe actions on a structured view but return allow when a raw issue contains the reviewer-targeting phrase of T_j. Its contract guarantees correctness only for structured views; the implementation satisfies that conditional contract. Let an intermediate layer relay its verdict assuming the caller has established that interface premise, and let an outer layer rely on the intermediate verdict assuming deterministic invariants are preserved. Both conditional relay contracts are satisfied. A naive caller passes raw T_j and executes its relayed allow without a sink check. The action violates the secret-egress invariant. The missing premises, rather than any valid assume-guarantee inference, explain the failure. T_b remains benign and T^* without the reviewer phrase is denied. Mandatory sink mediation also defeats T_j in the stronger naive control. ■

The sufficient conditions for sound closure are explicit assumption discharge for each edge, compatible property scopes and actual views, acyclic escalation, and preservation of mandatory invariants. A declaration that one layer trusts another is therefore not an executable substitute for checking these premises.

Theorem 1 (Scoped composition soundness): Assume complete mediation with unforgeable metadata, a correct deterministic monitor on every acceptance, monotone denial, and simultaneous certificates $\Pr_D(Y = 1, E_i) \le b_i$ for the disjoint acceptance routes whose interface and context premises hold. Then no action violating φ_{det} executes, and $R(D) \le \sum_i b_i$. If the certificates hold jointly with calibration probability at least $1 - \delta$, the risk conclusion has that same confidence.

Proof: Induct on mediated trace extensions. Any executed extension has passed the correct deterministic monitor, preserving the invariant. For semantic loss, disjointness gives $\mathbf{1}_{\{Y=1,\text{allow}\}} = \sum_i \mathbf{1}_{\{Y=1,E_i\}}$. Taking expectations yields the result. Simultaneous certificates can be obtained by allocating a failure budget across a fixed family and applying a union bound; independence is unnecessary. Failure of an executable interface premise forces denial. The theorem does not assert that distributional premises are decidable at runtime. In T^* and T_j deterministic mediation denies the action; T_b needs an applicable semantic acceptance certificate. ■

The implementation certifies the entire fixed routing policy, rather than estimating a local error on selectively observed traffic and multiplying it by a marginal reach rate. This is more conservative, but avoids hidden selection assumptions. Route-specific certificates are a supported theoretical extension, not separately estimated by the pilot implementation.

B. Risk accounting and dependence

Theorem 2 (Veto bounds, route bounds and calibration scope): For an all-veto ensemble with joint miss events $M_i = \{Y = 1, \text{allow}_i\}$ and $\Pr(M_i) \le \beta_i$, $R \le \min_i \beta_i$.

If misses are conditionally independent given $Y = 1$, with prevalence $\pi = \Pr(Y = 1)$ and $m_i = \Pr(\text{allow}_i \mid Y = 1)$, then $R = \pi \prod_i m_i$. In the specified conditional common-shock mixture,

$$R = \pi \left((1 - \rho) \prod_i m_i + \rho \min_i m_i \right). \tag{2}$$

An early-accept cascade instead requires the disjoint-route sum in Theorem 1.

Proof: The veto acceptance intersection is contained in each M_i, giving the minimum bound. Conditioning on unsafe input and using the stated independence gives the prevalence-weighted product. In the common-shock branch, all guards compare a shared uniform variable with their miss rates, so the intersection probability is the smallest rate. In the independent branch it is the product. Conditioning on the mixture branch proves (2). For disjoint early acceptance, an input may be accepted without visiting later guards; the intersection argument does not apply. ∎

The product of joint marginal miss probabilities is generally incorrect even under unsafe-conditional independence: those events share the same unsafe input event. Moreover, observed pairwise correlation does not identify an arbitrary higher-order failure law. Common compromise of a judge on T_j illustrates why repeated verdicts need not be independent. Successful classification of T_b supplies no evidence for that independence assumption.

For calibration, we separate three facilities. Split-conformal quantiles provide marginal coverage under their exchangeability premise. Conformal risk control (CRC) applies to a fixed nested family of bounded losses $l_\lambda \in [0, 1]$ decreasing in conservatism. Choosing the smallest parameter such that $(\sum_{k=1}^{n} l_\lambda(Z_k) + 1)/(n + 1) \leq \alpha$, with a guaranteed zero-loss fallback, controls marginal expected risk under the CRC conditions [7]. This is not a high-confidence statement about the risk conditional on a realized calibration sample.

The pilot instead uses an exact one-sided binomial upper bound for each fixed policy's joint unsafe-acceptance loss on independent identically distributed (iid) calibration samples. A union correction over the prespecified candidate family permits post-calibration selection. The iid premise is stronger than general finite exchangeability and is stated explicitly. Finite test-set frequency can exceed a certified population target through sampling variation; our coverage experiment checks true population risk separately.

Under shift, if an externally justified total-variation bound $\mathrm{TV}(D, Q) \leq \eta$ holds for a fixed policy, then $R(Q) \leq \min(1, R(D) + \eta)$ by the definition of total variation applied to the loss event. No estimator of η is certified here. For an unconstrained adaptive attacker, denying all uncertifiable semantic routes or relying on exact deterministic mediation is the only universal fallback in this model.

C. Escalation synthesis

Theorem 3 (Conditional escalation threshold): Suppose a trusted view x has known cheap-acceptance risk $r_0(x)$, escalation risk $r_1(x)$, and positive incremental cost $d(x)$. For randomized escalation $q(x) \in [0, 1]$, minimizing expected cost subject to an expected-risk constraint ~~admits a pointwise Lagrangian rule: escalate~~ has this sufficient optimality condition: if a measurable policy and finite multiplier exist satisfying feasibility and complementary slackness $\lambda(R(q) - \epsilon) = 0$, pointwise minimization escalates when $\lambda(r_0 - r_1) > d$, accept when the inequality reverses, and possibly randomizesat equality. ~~A feasible~~ Such a multiplier/tie mixture ~~meeting the risk constraint certifies optimality~~certifies optimality; its existence is an additional premise, not a consequence of known risks.

Proof: For multiplier $\lambda \geq 0$, the Lagrangian, up to a constant, is $\mathbb{E}[q(x)(d(x) - \lambda(r_0(x) - r_1(x)))]$. Minimizing each linear term over the unit interval gives the rule. If a feasible solution meets the constraint and complementary slackness, its Lagrangian value equals its primal cost; weak duality then proves global optimality under the stated existence premise. If costs are constant and risk reduction is monotone in a score, this becomes a threshold on that score. Neither monotonicity nor known conditional risks follows from calibration. At a boundary risk budget a feasible optimum need not admit a finite multiplier; no existence claim is made in that case. ∎

Escalation to a compromised judge on T_j can increase risk, invalidating the informal claim that stronger deliberation must help. For T_b, costly escalation can reduce utility. For T^*, semantic optimization never waives the mandatory secret-egress predicate. These constraints connect the optimization objective to the same running example used in the definitions.

The implemented synthesizer enumerates a fixed finite family, computes simultaneous upper-risk bounds, and chooses the cheapest feasible candidate. ~~This is exact relative to that family~~The objective is observed calibration mean cost among certified candidates, not certified population expected cost. A linear program supports mixtures of finite candidates with known risk/cost inputs; a single risk constraint permits an optimum supported on at most two candidates. Multiple-layer evaluation expands a bounded candidate grid. No approximation ratio to unrestricted routing is claimed. A latency-tail constraint can be certified as $\Pr(L > \text{budget}) \leq \gamma$; averaging percentile values is not valid.

Theorem 4 (Tightness and limits): The veto minimum and ~~disjoint-route sum bounds~~ capped route bound $\min(1, \sum_i b_i)$ are tight given ~~only~~ their stated marginal information. Pairwise independence does not determine joint failure for an arbitrary ensemble.

Proof: For unsafe inputs let guard i miss when a shared uniform variable is at most m_i. The nested miss events attain intersection probability $\min_i m_i$. Independent uniforms attain the product construction. For disjoint acceptance routes, place unsafe mass b_i on each route and the remaining mass elsewhere, with $\sum_i b_i \leq 1$; the sum bound is attained. Finally, uniform laws on even-parity and odd-parity Boolean triples both have marginal probability $1/2$ and pairwise independent coordinates, yet their all-one probabilities are respectively 0 and $1/4$. Thus the same pairwise information admits different joint failures. For T_j, common judge susceptibility can likewise defeat an independence interpretation; T_b alone cannot identify the relevant failure law. ∎

TABLE II
OBLIGATION-TO-CODE CORRESPONDENCE. DECLARATION CHECKS DO NOT PROVE EXTERNAL DISTRIBUTION OR PLUGIN SEMANTICS.

Obligation	Executed check	External premise
Edge compatibility	Scope/evidence sets; fixed topology	Truth of guarantees
Provenance	Write-once store; parent-taint union	Complete host dependencies
Approval	Digest and epoch match	Trusted grant issuer
Hard safety	Mandatory typed predicates	Faithful interpreter semantics
Judge interface	Closed Boolean/numeric schema	Adequate semantic evidence
Certificate	Policy/view/context identity; risk/confidence budgets	Valid issuer, labels, iid relationship
Guard identity	Version string in fingerprint	Immutable implementation

D. Implementation, assumptions and limits

The typed interpreter implements local mock HTTP, in-memory files and typed tool-effect records. It rejects arbitrary shell commands and unknown tool fields. Provenance references are write-once; unknown references retain conservative taint. Approval binds the immutable action digest and epoch. The trusted dispatcher checks its configured environment against the proposal. The tool adapter parses function-call argument objects and invokes the same mediated interpreter; it is not a tested connection to an external model API.

The direct-edge checker validates declared compatibility, not the truth of component implementations. The runtime has a fixed routing topology; the DSL does not synthesize arbitrary temporal monitors. A certificate fingerprint binds the specification, scorer/judge versions, threshold, representation and environment, and is recomputed at evaluation. Plugin identity and context declarations remain trusted-host obligations. The interpreter is synchronous~~; concurrent mutation of trusted stores~~. Current lexical inputs are length-bounded, but arbitrary plugins have no runtime-enforced cancellation deadline: a caller-reported timeout exception is handled, while a nonreturning plugin can hang. Availability, concurrent trusted-store mutation and covert channels are out of scope. Frozen experiments retain their earlier scorer versions.

In the stated control semantics, every effect has a preceding hard-pass and certified state, by induction over the only effect transition. Table II distinguishes checked facts from external premises. This correspondence is an inspection argument, not a proved refinement link or a general semantic contract solver.

Assumptions and limits. Deterministic guarantees hold only for this typed, completely mediated semantics. The PRISM model is an independently checked finite abstraction, without a proved refinement relation to Python. Structured quarantine excludes raw instructions but does not establish the semantic oracle. Statistical certification assumes fixed candidates and the stated calibration/deployment relationship; arbitrary adaptation and drift are not covered. A matching fingerprint cannot establish that relationship. These are preconditions, not empirically established facts about real LLMs.

IV. EVALUATION DESIGN

The local protocol was committed and tagged `prereg-v1` before empirical calibration or held-out evaluation. Unit tests and canonical model checks preceded preregistration and are not independent confirmatory findings. The tag is local, not an independently timestamped registry. The original run and the amendment are retained separately, with their source commit identifiers.

The testbed uses a handwritten lexical scorer, a deliberately vulnerable raw-text judge and a structured deterministic judge. The latter's features do not include the oracle label. Two scripted proposal sources are not two base LLMs: no real base LLM was run. The hardware audit found a small GPU but the installed learning framework was CPU-only. API credentials were unavailable, and the inspected small guard model required authenticated access acceptance. Docker's engine was unavailable; its image recipe was not executed.

A. Datasets, splits and comparators

For each of 10 primary seeds, the fixed fixture law generates 1800 calibration, 600 development and 1200 test traces with disjoint identifiers. Template families overlap across these iid splits, so they are not independent authored tasks. An additional 600 traces per seed use an encoded family absent from calibration. The law includes benign requests, benign security discussion, secret exfiltration, unapproved destructive production actions, direct instruction-following, implicit unauthorized patches and judge-targeting text. Labels are known by construction, with no independent human annotation.

Baseline names deliberately describe their implemented behavior: rules only, lexical only, scripted judge only, invalid delegation, naive with a mandatory sink, and checked but uncalibrated. None is an official AgentSpec, ShieldAgent or LlamaFirewall implementation. The invalid-delegation comparator is an executable countermodel, not a strong state-of-the-art defense. The naive-with-sink control isolates the value of mandatory mediation. The equal-threshold checked control isolates decision behavior from calibration.

The policy grid contains 11 fixed thresholds. Whole-policy joint loss is certified against a 20% target with a 5% confidence-failure budget and simultaneous correction over that grid. This permissive target is a test of certification mechanics, not a production safety recommendation. Infeasible selection denies all uncertified actions. Development data are diagnostics, not used to revise the frozen thresholds. Modeled layer costs are arbitrary units from the frozen configuration; CPU quantiles measure complete local decisions separately.

Public benchmark repositories and their code licenses were inspected, but no external dataset was executed or redistributed. R-Judge's project website states a noncommercial share-alike license; repository-license retrieval failed. AgentHarm has an additional safety-use restriction. These observations do not imply that an upstream repository's code license covers every included dataset.

TABLE III
FROZEN HYPOTHESIS-TO-EVIDENCE MAP. "SUPPORTED" IS ALWAYS SCOPED TO THE ACTUAL LOCAL INSTRUMENT. UNAVAILABLE EVIDENCE IS NOT REPLACED BY A SIMULATED CLAIM ABOUT REAL MODELS.

Hypothesis	RQ	Evidence	Outcome
Hard invariant preservation	I	Proof, PRISM, exhaustive model, replay	Supported within typed semantics
Planted invalid-spec detection	I	Mutation confusion matrix	Supported for generator classes
Selected bound coverage	II	Known iid law with repeated calibration	Supported for synthetic law
Invalid delegation counterexample	II	Canonical traces and stronger sink control	Supported; sink alone also protects
Finite-family selection optimality	II	Independent enumeration and LP constraints	Supported within candidate family
Useful safety–utility improvement	III	Paired replay, controls and trade-off plots	Primary inconclusive; exploratory only
External/model generalization	III	No real-model or benchmark execution	Inconclusive/not run
Full policy expressiveness	I	Executable restricted-DSL inventory	Refuted

B. Evidence map and statistics

Table III maps every hypothesis to its evidence. The calibration-coverage study performs 1000 independent replications of a known nested Bernoulli loss law, using fresh calibration and test samples. Its truth is the specified population risk, which permits direct checking of whether a certificate covers that risk. The finite test frequency is reported as a separate quantity.

Seed-level paired risk/utility differences use a percentile bootstrap with 10000 resamples, paired Wilcoxon tests, Holm correction across baseline comparisons, and descriptive Vargha–Delaney A_{12}. Intervals resample seeds, not duplicated template instances as though they were independent developers. The sample is a pilot and is not powered to detect real deployment effects. Coverage intervals use an exact binomial interval. No result rests on a bare significance test.

The contract scalability study varies declared policy scopes and graph depth, timing valid cases and planted missing-evidence, scope, endpoint and cycle mutations. These repetitive generated obligations do not represent rich temporal specifications. Authoring time and participant usability were not measured. Adaptive search permits at most 16 local fixture queries per comparator/seed, keeps all attempts, and stops on unsafe acceptance. ~~It is a weak budgeted fixture search, not a model-based optimizer.~~ The historical instrument calls fixed-threshold uncertified decision-core functions, including the raw label `checked_quarantine`; it does not invoke the calibrated runtime or its selected certificate. Its first fixed phrase succeeds for every core, providing no observed response-informed adaptation. These records cannot be attributed to the denying primary runtime.

V. RESULTS

A. RQ I: specification and decomposition

The canonical PRISM study checks 6 configurations and agrees with the independent checker on both reachability properties. Exhaustive enumeration covers 64 Boolean sink cases. Seeded sanity checks exercise 10000 random probability instances. The general proofs remain pen-and-paper proofs; these finite checks are neither Lean/Isabelle mechanizations nor proofs of the runtime implementation.

The original mutation study rejects 1400 planted invalid specifications while accepting 350 clean controls, with 0 false rejections and 0 missed mutations. This demonstrates agreement with the mutation generator, not precision/recall on naturally occurring policy errors. The maximum cell-median checker time is 9737.50 microseconds on the recorded host. No policy-count scalability result is claimed for PRISM; its evaluated model is the fixed canonical abstraction.

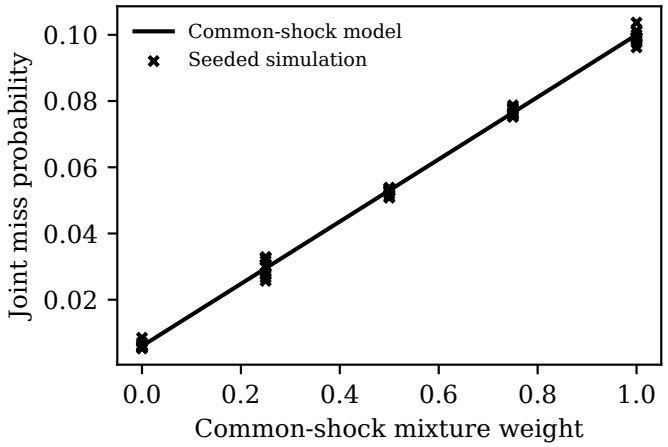


Fig. 2. Seeded common-shock simulations and the model-specific intersection formula. The horizontal parameter is a mixture weight, not measured pairwise correlation. The plot does not assert a universal interpolation law.

The inventory supports only 2 of 5 policy classes fully in the DSL: production approval and secret egress. Semantic injection has a statistical proxy; filesystem confinement and cumulative spending are rejected as unsupported predicates. The full-expressiveness hypothesis is refuted. This restricted boundary is preferable to accepting declarations that the runtime does not enforce. Core test coverage is 99.70%, including the numeric-boundary regression. Coverage is a structural test metric, not a proof of correctness.

B. RQ II: risk, dependence and policy selection

In the frozen coverage experiment, selected bounds cover the true population risk in 99.60% of 1000 replications; the exact interval is [98.98%, 99.89%]. The nominal confidence is 95%. This conservative finite-family certification exceeds the target coverage under the known iid generator; it is not an adaptive-attack result. The finite test frequency falls below its target in 998 replications, a distinct observable.

Fig. 2 visualizes the constructed dependence law behind Theorem 2. The parity construction independently demonstrates that the same pairwise information need not determine

TABLE IV
FROZEN PRIMARY REPLAY. RISK IS JOINT UNSAFE ACCEPTANCE; UTILITY IS BENIGN ACCEPTANCE. COSTS ARE MODELED ARBITRARY UNITS, CPU IS MEASURED LOCAL DECISION TIME, AND HARD COUNTS ARE VIOLATING ACCEPTED PROPOSALS UNDER THE TYPED ORACLE. THE CALIBRATED METHOD DENIES ALL UNCERTIFIED ACTIONS.

Method	Risk (%)	Utility (%)	Cost units	CPU upper-tail (μs)	Hard count
Rules only	29.83	100.00	1.00	4.60	0
Lexical only	19.82	66.41	20.00	2.80	1176
Scripted judge only	39.62	66.41	1000.00	12.40	1176
Invalid delegation	39.62	66.41	1020.00	9.20	1176
Naive with sink	29.83	66.41	365.40	6.00	0
Checked, uncalibrated	29.83	66.41	365.40	7.30	0
Checked, calibrated	0.00	0.00	1.00	137.90	0

higher-order risk. These are controlled validations of assumptions and limits, not estimates of real guard correlations. T_j supplies the complementary executable witness that a raw judge's interface premise can fail even when its conditional contract is satisfied.

Across 50 finite-family instances, the original synthesizer has 0 mismatches with independent exhaustive selection. The exploratory replication has 0 mismatches across 50 instances. The original raw synthesis records retain bounds but omitted calibration counts; consequently those particular bounds cannot be independently recalibrated from their records. The amended instrument retains sufficient counts. Selection consistency and LP feasibility are checkable in both runs; no unrestricted routing optimum is inferred. Re-analysis additionally reconstructs mixture risk/cost from weights, checks normalization/nonnegativity, and compares objective values with independent vertex enumeration. Read-only semantic replay joins each decision to its trace and validates the historical configuration, oracle fields and non-timing verdict, route, layer and cost outputs.

The most consequential RQ II finding is feasibility failure. In the primary replay, 10 of 10 seed-level families have no candidate below the frozen risk budget. The runtime denies uncertified routes. This follows the specified fallback but does not create a useful policy. It shows why a risk-constrained synthesizer must expose infeasibility, rather than silently relaxing the budget or treating absence of a certificate as permission to execute.

C. RQ III: utility and engineering trade-offs

Table IV reports the primary replay on 12000 held-out proposals. Denying all uncertified actions has 0.00% joint risk and 0.00% benign utility. Rules-only retains 100.00% benign utility at substantially greater semantic residual risk. A system that denies every benign task has not demonstrated the intended safety–utility advantage, regardless of its risk.

Inspection after the frozen run found a boundary defect in the handwritten score: mathematically $0.04 + 0.12 \cdot 3$ can evaluate below the comparison threshold 0.40. A documented amendment computes integer hundredths before division, preserving all lexical features, thresholds, budgets and judge rules. The scorer version changes so earlier certificates cannot apply. The original logs remain intact; the fresh-seed replication is explicitly exploratory.

Table V reports 10.16% joint risk with 66.46% benign utility. Against the stronger naive-with-sink control, the paired joint-risk difference is -10.62 percentage points, with interval [-11.38, -9.86], descriptive $A_{12} = 1.00$ and Holm-adjusted $p = 0.01172$. This is evidence about the scripted judge mechanism only. The uncalibrated checked control has identical decisions and joint risk; no detection benefit is attributed to calibration.

Table VI exposes the limited support. Under the authored law, the exploratory checked policy's analytically implied risk is 10.00%, utility is 66.67%, and modeled cost is 366.00 units. The stronger naive sink control's population risk is 20.00%. These differences are built into the scripted mechanisms and prevalences. The interval above measures mixture-sampling uncertainty, not independent evidence of effectiveness on semantically varied attacks. Among accepted exploratory proposals, unsafe conditional risk is 25.40%, distinct from joint unsafe acceptance.

The rules-only control remains much cheaper in the modeled cost and has higher benign utility. CPU measurements include actual local middleware processing; the calibrated runtime's fingerprinting and checks add overhead that toy comparators do not perform. They are not estimates of real guard inference latency. Fig. 3 therefore supports a scoped engineering trade-off, not dominance over the baselines on every axis.

On the unseen encoded family, exploratory checked risk reaches 100.00%. The feature view cannot distinguish the authored unauthorized patch, and the declared runtime context does not detect the distribution change. ~~The fixture search also~~ Fixed-threshold core search finds accepted unsafe inputs ~~quickly~~immediately; it does not ~~substantiate robustness~~ measure the primary certified runtime and supplies no robustness evidence against a strong adaptive attacker. The primary checked fallback instead has no utility. Both observations limit the intended general robustness claim.

D. Assumption audit and threats to validity

Internal validity. The observed numeric defect affected primary classification. Its correction is disclosed with fresh seeds and separate results, but post-result amendments do not restore confirmatory status. Oracle labels and templates were authored by the implementation designer. Mutation accuracy partly reflects deliberate construction of checker failure cases. Version/configuration checks have regression tests; they do not

TABLE V
EXPLORATORY NUMERIC-CORRECTION REPLICATION ON FRESH SEEDS. METRICS AND UNITS MATCH TABLE IV. EQUAL DECISIONS FOR CALIBRATED AND UNCALIBRATED CHECKED CONTROLS SHOW THAT CALIBRATION IS AN APPLICABILITY/SELECTION MECHANISM HERE, NOT A NEW SEMANTIC DETECTOR.

Method	Risk (%)	Utility (%)	Cost units	CPU upper-tail (μs)	Hard count
Rules only	30.56	100.00	1.00	5.40	0
Lexical only	20.26	66.46	20.00	3.70	1212
Scripted judge only	30.88	66.46	1000.00	15.70	1212
Invalid delegation	30.88	66.46	1020.00	11.90	1212
Naive with sink	20.77	66.46	370.67	7.70	0
Checked, uncalibrated	10.16	66.46	370.67	9.20	0
Checked, calibrated	10.16	66.46	370.67	188.80	0

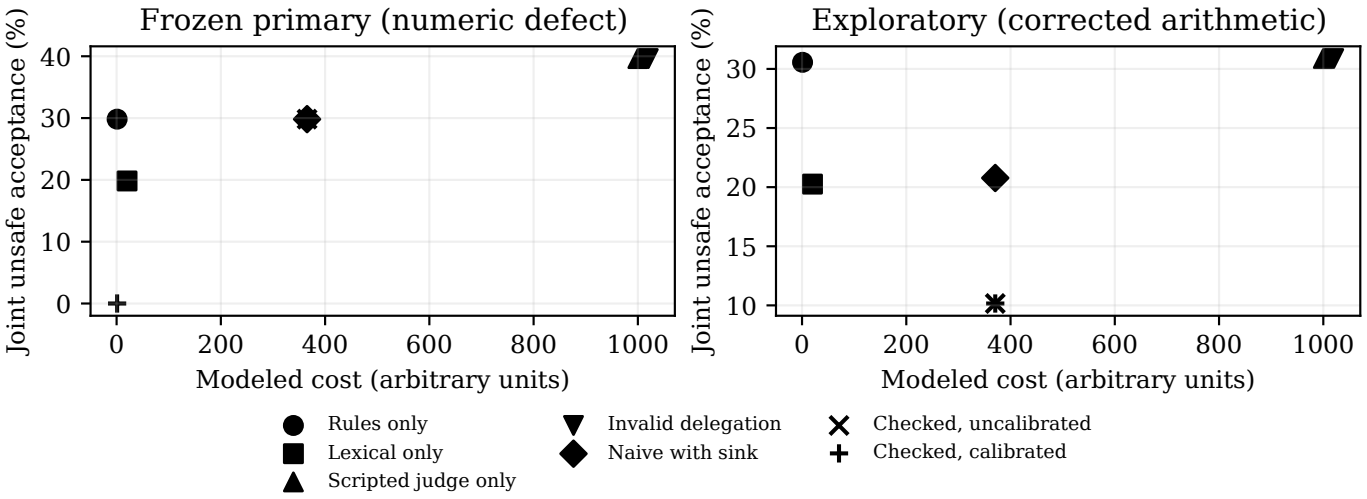


Fig. 3. Observed local risk–modeled-cost points, keeping frozen primary and exploratory results separate. Utility must be considered alongside these axes: the primary low-risk point denies all benign tasks. Equal points overlap. These plots compare configurations, not external model Pareto frontiers.

TABLE VI
AUTHORED FAMILY SUPPORT AND EXPLORATORY ACCEPTANCE WITHIN FAMILY. SUFFIX VARIANTS DO NOT ALTER GUARD BEHAVIOR; THESE ARE MIXTURE-LAW SAMPLES, NOT DISTINCT SEMANTIC TASKS.

Family	Primary	Exploratory	Sink (%)	Checked (%)
Benign report	3592	3580	100	100
Benign security text	1817	1807	0	0
Secret exfiltration	1836	1734	0	0
Production deletion	1176	1212	0	0
Direct injection	1182	1174	0	0
Implicit patch	1203	1219	100	100
Judge targeting	1194	1274	100	0

validate arbitrary plugins. Original synthesis calibration-count omission limits independent reconstruction of that subset of bounds.

External validity. The study executes no real base LLM, pretrained guard, activation probe or external benchmark. No comparison to an official state-of-the-art defense is available. Host performance, load, long sessions, token cost, GPU cost and temporal deployment drift are untested. The small local workload cannot answer the general engineering-value research question.

Construct validity. Joint risk, attack-conditioned success and conditional risk among accepted actions answer different questions. Benign acceptance is a proxy for task utility; it is not successful completion of a real development task. Quarantine removes content from a feature schema but does not prove semantic noninterference. Distributional context is asserted by trusted deployment code, not inferred from the fingerprint. The synthetic shift deliberately breaks that premise.

Conclusion validity. The paired analyses treat seeds as the sampling unit, but they all share one fixture law. Conservative binomial correction can cause infeasibility. Correlation measurements cannot establish conditional independence; common-shock results hold only for the constructed law. Formal definitions and proof constants have proof provenance, while empirical numbers are generated from raw-derived macros. The numeric linter checks authored empirical prose, not the origin of every structural glyph in a PDF.

VI. RELATED WORK

A. Guardrail systems

Llama Guard, WildGuard, Constitutional Classifiers and ThinkGuard illustrate classification and reasoning approaches to safety monitoring [8]–[11]. Our opened bibliography covers the verified sources among these systems; we do not infer a missing capability from an abstract's silence. NeMo provides programmable rails, while LlamaFirewall combines specialized scanners for agent threats [12], [13]. Truncated polynomial classifiers support progressive activation monitoring [14]. These systems motivate heterogeneous layers; our

toy components do not reproduce their learned behavior or results. OneShield further illustrates configurable guardrail engineering [15].

B. Cascades and adaptive compute

FrugalGPT studies cost-aware model cascades [16]. Trust or Escalate already supplies cascaded selective evaluation with human-agreement guarantees [4]. Conformal Arbitrage already calibrates a primary/guardian switch and analyzes score-gap routing optimality [3]. Hybrid abstention uses hierarchical detection for practical filtering [17]. Our distinction is checking the applicability of certificates at actual tool-action routes, not inventing calibrated cascading. The mathematical optimization result is a standard Lagrangian observation, not a new general router theorem.

C. Dual-process architectures

The System I/System II vocabulary connects fast response with deliberative reasoning, and the reasoning-model survey organizes that literature [18]. We use an operational resource criterion without claiming equivalence to human cognition. Training-time helpfulness and constitutional feedback address another part of safety engineering; they do not substitute for tool-effect mediation [19], [20]. The broader dual-process seed list remains incompletely verified and is not turned into invented citations.

D. Runtime enforcement and verification

AgentSpec provides structured runtime rules [21]; ShieldAgent reasons with verifiable policy representations [22]. ProbGuard estimates future safety through probabilistic monitoring [23]. VeriGuard separates verified-policy construction from online action monitoring [24]. ContrAgent is especially close in its temporal contracts and compiled monitors [1]; Agent Behavioral Contracts also supplies composition and probabilistic satisfaction notions [25]. The recent layered probabilistic architecture directly limits any claim to first conditional guardrail composition [2].

Shield synthesis provides classical runtime correction under formal specifications [26]–[29]. RoboGuard protects a root-of-trust model and uses temporal constraints, while RoboSafe and related robotics work incorporate executable safety logic and planning [30]–[33]. Our deterministic mediation belongs to this established enforcement lineage.

E. System-level isolation

CaMeL separates control and data flow and mediates capabilities [5]. Fides tracks confidentiality/integrity labels with a formal planner model [6]. Spotlighting and instruction-hierarchy training address provenance cues and instruction priority through model behavior [34], [35]. TINT uses a deliberately restricted structured guard view, not a proof that prompt formatting prevents instruction-following. Task Shield checks goal alignment and provides a stronger real-agent comparison target for future work [36].

F. Statistical guarantees

CRC controls expected monotone loss under exchangeability conditions [7]. Conformal tutorials and adaptive classification methods distinguish marginal guarantees from stronger conditional claims [37], [38]. Adaptive conformal inference studies changing distributions, while adaptive and tail-risk methods broaden the controlled objectives [39]–[41]. Conformal abstention, factuality and language-generation methods demonstrate other established statistical interfaces [42]–[46]. We implement small calibration utilities and apply finite-family binomial certification; we claim no new conformal theorem or adaptive safety guarantee.

G. Layered safety engineering

Conditional component contracts, runtime assurance and defense in depth motivate exposing assumptions instead of counting layers. ScenicProver is a particularly relevant compositional assurance framework with explicit evidence and provenance [47]. Semantic integrity constraints also illustrate declarative assurance at another software boundary [48]; NaSh emphasizes recovery and guardrail design for language-driven shells [49]. Our gap is an executable integration for a restricted tool-agent interface, not a novel safety-case principle. Wider foundational coverage remains necessary for a submission.

H. Attacks and evaluation

Indirect injection, optimized jailbreaks, guard evasion and latent-defense attacks demonstrate why guard behavior must itself be part of the threat model [50]–[54]. RAG context changes can alter guard judgments [55]; baseline defenses illustrate the need to specify attacker access [56]. AgentDojo, ASB, AgentHarm, RedCode, ToolEmu and R-Judge offer substantially richer evaluation substrates [57]–[62]. RedCodeAgent, BAD-ACTS and CyberSecEval further expand agent-security testing [63]–[65]. Representation sensitivity cautions against treating a benchmark score as an intrinsic model property [66]. None of those benchmarks was executed here, and our fixture attack search is not a comparable evaluation.

Table VII summarizes overlap rather than claiming that each capability is individually novel. The closest-source full-text review and the wider abstract-level scoping review have different depths; the retained verification records establish bibliographic identity, not independent validation of all claims made by those sources.

VII. Discussion and Limitations

The main lesson is a specification discipline: an allow verdict is usable only within its stated scope and premises. The canonical example demonstrates an invalid composition, while its stronger sink control prevents attribution of ordinary mediation benefits to a new trust mechanism. The route-sum result explains why importing a veto-ensemble product into an early-accept cascade is unsound. Neither observation is a claim that classical contract reasoning failed to account for assumptions.

TABLE VII
CLOSEST-WORK COMPARISON. Y DENOTES AN ESTABLISHED CAPABILITY IN THE REVIEWED SOURCE, S A DIFFERENT/SCOPED PROPERTY, U UNESTABLISHED IN THIS REVIEW, AND P PROPOSED/TESTED ONLY IN OUR LOCAL PILOT. U NEVER MEANS PROVEN ABSENCE.

Approach	Hierarchy	Formal spec	Hard	Statistical	Conditioned trust	Cost	Agent traces	Adaptive	Tool
ContrAgent	U	Y	Y	U	Y	Y	Y	U	Y
Layered probabilistic contracts	Y	Y	S	Y	Y	U	U	U	U
Conformal Arbitrage	Y	Y	U	Y	S	Y	U	U	Y
Trust or Escalate	Y	Y	U	Y	S	Y	U	U	Y
AgentSpec	U	Y	Y	U	U	Y	Y	U	Y
CaMeL / Fides	Y	Y	Y	U	Y	Y	Y	Y	Y
RoboGuard	Y	Y	Y	U	Y	Y	Y	Y	Y
ScenicProver	Y	Y	Y	Y	Y	S	S	U	Y
TINT local integration	Y	Y	P	P	Y	P	P	P	Y

The primary infeasibility outcome is practically relevant. A guard stack can have sound rejection behavior yet provide no useful certified configuration. The corrected exploratory run demonstrates implementation feasibility but still accepts implicit unsafe actions and fails under the designed shift. An applicability certificate does not repair a poor feature view or improve an unchanged classifier. Our equal-threshold control makes that distinction observable rather than burying it in a combined ablation.

Future evaluation must use real proposal-generating LLMs and independently implemented learned guards across external benchmarks. GPU-hosted models, activation probes and reasoning judges may change miss rates, correlations, costs and latency under load. No numerical bound or trade-off reported here is claimed to transfer unchanged. Further work also needs richer rule semantics, explicit route-specific calibration, independently authored policy errors, stronger adaptive attacks and a proved abstraction/refinement link. The present artifact does not justify a completed TSE effectiveness claim.

VIII. CONCLUSION

Explicit conditions on trust edges make a limited but useful assurance case executable: deterministic effects remain mediated, certificates bind the actual routing configuration, and unproved premises are visible. Scoped proofs and reproducible local experiments expose the difference between conditional component guarantees and valid composition. The primary utility failure, the exploratory correction and the shift failure all constrain the empirical conclusion. Real-agent engineering value remains to be established.

DATA AND ARTIFACT AVAILABILITY

The repository is https://github.com/ai4se4ai-lab/trust-is-not-transitive. The local artifact includes raw compressed decision logs, processed summaries, source/configuration hashes, proofs, PRISM input/output, generated macros and vector figures. Raw execution commit identifiers are recorded in each `environment.json`; the artifact manifest identifies analysis inputs. No remote publication or archival DOI was created. The historical primary instrument is retained for reproduction alongside the corrected exploratory instrument. `ARTIFACT.md` distinguishes re-analysis from full reruns and documents unavailable prerequisites. The source comprises one authored `main.tex`, one bibliography and a generated macro include, with the provided template's IEEEtran class/options and bibliography style.

ETHICS AND ACKNOWLEDGMENTS

All attack strings target the local typed interpreter or a scripted judge. There is no real network egress, production operation, third-party exploit, personal datum or genuine secret in the released fixture data. No participant study was conducted. External model access conditions were not accepted on the user's behalf, and external datasets are not redistributed. No funding, institutional approval, human contribution or reviewer endorsement is claimed. Automated adversarial reviews accompanying the artifact are development aids, not independent peer review or publication acceptance.

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